mbusgw-t2r

User Guide

Document Number: DO-T2RUG-EN-01A
Publication Date: August 2026
Revision Update Information: This is a new manual.
Operating System: Linux (glibc), kernel 4.x or later
Software Version: mbusgw-t2r X.00-11

NaPi World & StarLet Squad collaboration

mbusgw-t2r User Guide

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Preface

1. About This Manual

This manual describes the configuration, the operation and the troubleshooting of mbusgw-t2r --- a gateway which translates MODBUS TCP requests coming from the network into MODBUS RTU transactions on a serial (RS-485 or RS-232) line, and returns the answers back.

2. Intended Audience

This manual is written for a reader with no prior MODBUS or Linux administration experience: every step is spelled out and nothing is assumed. An experienced engineer may go straight to Chapter 3 (the settings file) and Chapter 7 (the troubleshooting).

3. Document Structure

Chapter 1 explains what the gateway does. Chapter 2 is the pre-flight checklist. Chapter 3 describes the settings file key by key. Chapter 4 covers the start, the command line options and the first end-to-end test. Chapter 5 describes the built-in Time Stamp pseudo device. Chapter 6 teaches how to read the log. Chapter 7 is the troubleshooting reference: the symptom, the cause, the action, and the reference of all the message codes.

4. Related Documents

README.md --- the installation sequence (the StarLet package, the build, `make install`). MODBUS over Serial Line Specification and Implementation Guide and MODBUS Application Protocol Specification --- the protocol itself, available from modbus.org.

5. Conventions

Convention	Meaning
`Monospace type`	Code examples, file names, commands and interactive screen displays.
`%T2R-E-CODE`	A message code as it appears in the gateway log; see Section 7.2.
`<...>`	A placeholder to be replaced by an actual value, e.g. a device name.
`$`	The shell prompt of an unprivileged user; do not type it.
`Ctrl/C`	Hold down the key labeled Ctrl while you press the key C.

Chapter 1. What the Gateway Does

You have a device (an electricity meter, a temperature controller, a PLC ...) which talks MODBUS RTU over a serial line. And you have a program (a SCADA, a script, `mbpoll` ...) which talks MODBUS TCP over the network. The gateway sits between them and translates, as Figure 1 shows.

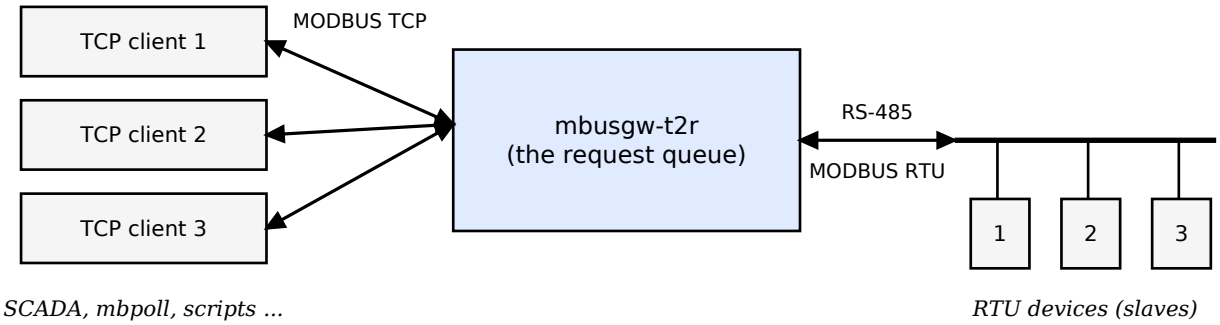


Figure 1. The data path of the gateway

Several TCP clients may connect at the same time: the gateway queues them so that only one request at a time goes to the serial line --- an RTU line physically cannot carry two transactions at once.

Several devices may share one RS-485 line. They are told apart by the slave address inside the MODBUS request itself; the gateway does not need to know their addresses at all.

Chapter 2. Before You Start

Walk this checklist once --- it saves an hour of guessing later.

1. The gateway is installed (see README.md). After the installation you have the program `/opt/mbusgw-t2r/sbin/mbusgw-t2r` and the settings file `/opt/mbusgw-t2r/etc/modbus-t2r_settings.conf`.

2. You know which serial port the device is connected to. On Linux it is a file like `/dev/ttyS0` or `/dev/ttyUSB0`. If unsure, plug the USB adapter out and in, then run:

```
$ dmesg | tail
```

3. You know the line parameters of the device: the speed, the data bits, the parity and the stop bits. They are in the device manual; the most common set is `9600, 8, N, 1`.

4. Your user may open the port. Check the group of the port file:

```
$ ls -l /dev/ttyUSB0
crw-rw---- 1 root dialout 188, 0 ... /dev/ttyUSB0
```

If the group is `dialout`, add yourself to it and re-login (or simply run the gateway with `sudo`):

```
$ sudo usermod -a -G dialout $USER
```

Chapter 3. The Settings File

The settings file has two sections: `serials` (the serial lines) and `listeners` (the TCP ports). A minimal working example:

```
serials = (
{ device = "/dev/ttyUSB0";
  chars  = "9600, 8, N, 1";
  iotmo  = 1000;
}
);

listeners = (
{ bind   = "TCP:0.0.0.0:502";
  target = "/dev/ttyUSB0";
  connlm = 8;
}
);
```

Read it as: "open the port /dev/ttyUSB0 at 9600-8-N-1 and wait for an answer up to 1000 ms; listen for TCP clients on every interface, port 502, and forward them to that port".

3.1. The serials Section

One record per serial port. Table 3-1 lists every key.

Key	Required	Meaning and the allowed values
`device`	yes	The serial port file, e.g. /dev/ttyUSB0.
`chars`	yes	The line parameters: speed, data bits, parity, stop bits. Speed 50..4000000; data bits 5..8; parity N (none), E (even), O (odd); stop bits 1..2.
`iotmo`	no	How long to wait for the answer of the device, milliseconds. Default 1000.
`rs485`	no	1 --- ask the kernel to drive the RS-485 direction control (only for the ports which support it). Default 0.
`ts_enabled`	no	1 --- enable the Time Stamp pseudo device (Chapter 5). Default 0.
`ts_unit`	no	The slave address the pseudo device answers on. Default 135.
`ts_fncode`	no	The function code it answers to. Default 4.
`ts_base_reg0`	no	The first register of the time stamp. Default 135.

If a record is wrong, the gateway skips it and says why --- with the allowed range right in the message:

```
%T2R-E-SERRANGE, [serial #00:</dev/ttyUSB0>] --- speed (baud) 31 is out of range [50..4000000]
```

3.2. The listeners Section

Key	Required	Meaning and the allowed values
`bind`	yes	Where to listen: TCP:<IP address>:<port>. The port is 1..65535; 0.0.0.0 means every interface. UDP is not supported.
`target`	yes	Which serial port to forward to. Must match a device from serials exactly, character by character.

Key	Required	Meaning and the allowed values
`connlm`	no	How many TCP clients may wait in the connection queue, e.g. 8.
`iotmo`	no	The network I/O timeout, milliseconds.

Port 502 is the standard MODBUS TCP port, but on Linux the ports below 1024 need the root privileges. If you do not want to run as root, take e.g. `5502` and point your client there. Several listeners may share one target: the gateway serializes their requests.

Chapter 4. Running the Gateway

By hand, in a terminal (the messages go straight to the screen):

```
$ /opt/mbusgw-t2r/sbin/mbusgw-t2r /settings=/opt/mbusgw-t2r/etc/modbus-t2r_settings.conf
```

Option	Meaning
`/settings=<file>`	The settings file (Chapter 3).
`/trace`	The verbose tracing: every request and answer is dumped in hex. Priceless at the first run.
`/logfile=<file>`	Write the log to a file instead of the screen.
`/logsize=<octets>`	Rotate the log file above this size.

A healthy start prints (shortened):

```
%T2R-I-REVISNF, Rev: T2R X.00-06/aarch64(built at ...) (REV: 00.06.00)
%T2R-I-DEVADDED, Added device #00 [</dev/ttyUSB0>, Chars: <9600, 8, N, 1>, ...] --- added
%T2R-I-LSNADDED, Added listener #00 [Target: </dev/ttyUSB0>, Net: <TCP:0.0.0.0:502>, ...] --- added
%T2R-S-DEVREADY, Device </dev/ttyUSB0> [9600 baud, answer tmo: 1000 msec, t3.5: 4010 usec] --- is r
%T2R-S-LSNRRDY, [#3] Listener 0.0.0.0:502 [Target: </dev/ttyUSB0>] --- is ready
```

Two lines matter most: DEVREADY (the port is open) and LSNRRDY (the TCP port is listening). If you see both, the gateway is up. Figure 2 shows what happens to every request after that.

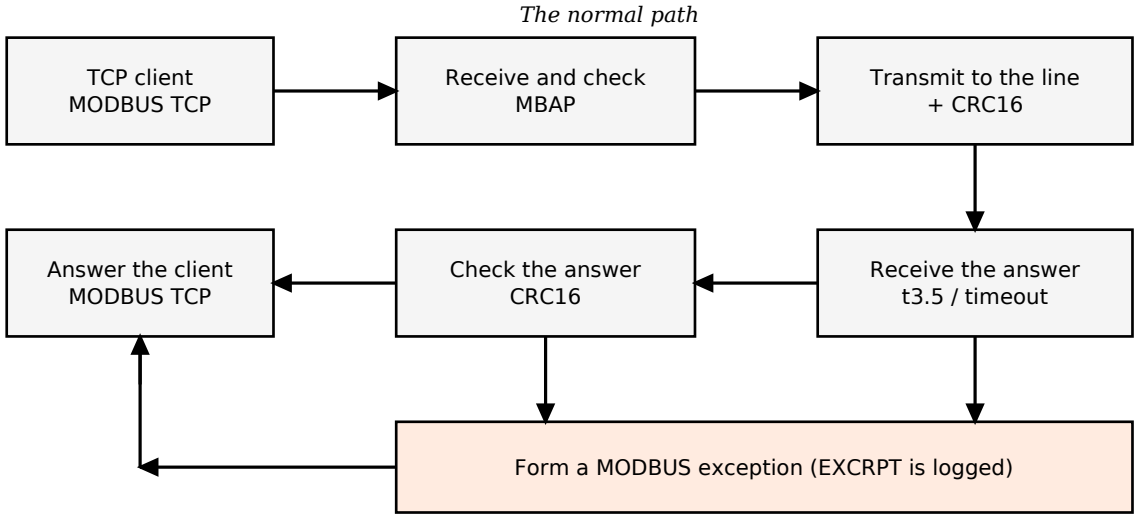


Figure 2. The lifecycle of one request

To stop the gateway press Ctrl/C once and wait a second. To toggle the tracing of a running gateway without a restart:

```
$ kill -USR1 <pid>
```

4.1. The First Test with mbpoll

`mbpoll` is a small free MODBUS client (`sudo apt install mbpoll`). Read 3 holding registers from the device with the slave address 1:

```
$ mbpoll -a 1 -r 1 -c 3 -1 127.0.0.1 -p 502
```

If the numbers come back --- everything works, end to end.

Chapter 5. The Time Stamp Pseudo Device

The gateway can answer by itself --- without touching the line --- to a read request sent to a special slave address. The answer is the current time of the gateway host. This lets a SCADA read the precise time over the plain MODBUS.

With `ts_enabled = 1; ts_unit = 135; ts_fncode = 4; ts_base_reg0 = 135;` a request "read N input registers starting at 135 from the slave 135" returns the registers of Figure 3.

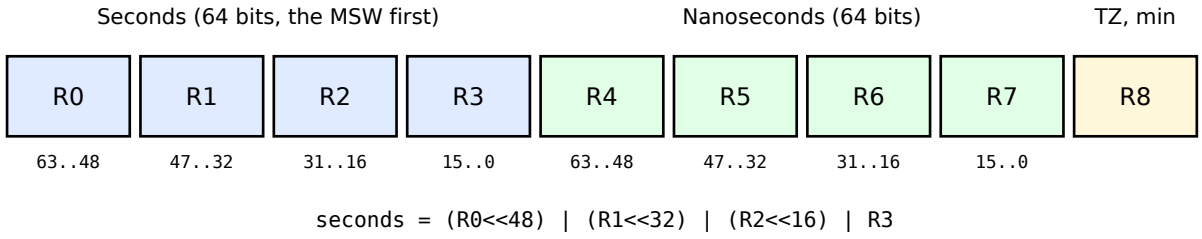


Figure 3. The Time Stamp registers layout

Registers requested	You get
4	The seconds of the UNIX epoch, 64 bits, the most significant 16-bit word first.
8	Plus the nanoseconds, packed the same way into R4..R7.
9	Plus R8 = the local timezone offset, in signed minutes (e.g. Moscow = +180).

Any other register count in a TS request is answered with the MODBUS exception ILLEGAL_DATA_ADDRESS --- this is on purpose.

Chapter 6. Reading the Log

Every important event is one line with a code:

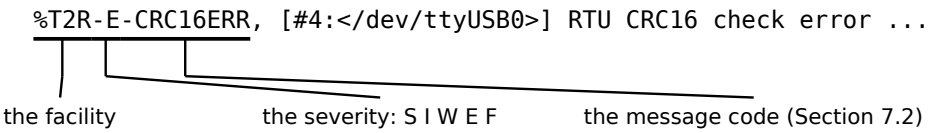


Figure 4. The anatomy of a log message

The severity letter is: S --- success, I --- information, W --- warning, E --- error, F --- fatal. The part `[#4:</dev/ttyUSB0>]` means: the file descriptor #4, this serial port.

Grep the log by the code --- this is exactly why the codes exist:

```
$ grep CRC16ERR /var/log/mbusgw-t2r.log
```

Chapter 7. Troubleshooting

The golden rule: run with /trace and read the log. The gateway always says what it dislikes.

7.1. The Symptom, the Cause, the Action

You see	It means / what to do
`%T2R-E-DEVOPNERR, ... errno: 2`	The port file does not exist: the adapter is unplugged or the name is wrong. Run dmesg | tail after plugging; fix device in the settings.
`%T2R-E-DEVOPNERR, ... errno: 13`	Permission denied. Run with sudo, or add yourself to the dialout group.
`%T2R-E-DEVOPNERR, ... errno: 16`	The port is busy. Find the holder: sudo fuser /dev/ttyUSB0, and stop it.
`%T2R-E-NOANSWER, ...`	The request went out, nothing came back: 1) a wrong slave address in the client request; 2) the wiring (swap A/B of RS-485); 3) a wrong speed/parity --- re-check chars; 4) the device is off.
`%T2R-E-CRC16ERR, ...`	The octets arrive damaged: 1) chars does not match the device (the most common!); 2) the line noise --- the grounding, the termination, the cable length; 3) two masters on one line.
`%T2R-E-BADFRAME, ...`	Something arrived, but it is not a sane MODBUS frame: usually the same causes as CRC16ERR; 400+ octets of garbage mean a foreign, non-MODBUS device on the line.
`CRC16ERR / BADFRAME now and then, on an FTDI USB adapter only`	The adapter gives the answer by pieces, once in 16 msec (latency_timer). Since X.00-10 the gateway waits for the whole answer of the functions 01-06, 15, 16, 23; the others are still cut. Set latency_timer to 1 msec (the answers get faster too): copy /opt/mbusgw-t2r/share/doc/mbusgw-t2r/99-ftdi-latency.rules to /etc/udev/rules.d/, run sudo udevadm control --reload && sudo udevadm trigger --subsystem-match=usb-serial; cat /sys/bus/usb-serial/devices/ttyUSB0/latency_timer must print 1.
`%T2R-W-FRAMETMO, ...`	The answer started but never finished: a very slow or hanging device. Raise iotmo; check the cable.
`%T2R-W-EXCRPT, ... (SERVER_DEVICE_FAILURE)`	The gateway told the client "the serial side failed". The real reason is in the previous log line (NOANSWER, CRC16ERR ...).
`%T2R-E-SESSFAIL, ... pthread_create()->11`	Too many TCP clients at once: every client takes a thread, and systemd limits the service (TasksMax, 578 on a small board). Find who keeps so many connections (ss -tn); if they are all real, set TasksMax=2048 in the [Service] section of the unit and run systemctl daemon-reload.
`%T2R-E-LSNRERR, ... errno: 98`	The TCP port is taken: a second gateway instance, or another program. Find it: sudo ss -tlnp | grep 502.
`%T2R-E-LSNRERR, ... errno: 13`	The ports below 1024 need root. Run with sudo, or take a port above 1024 (e.g. 5502).
`%T2R-E-SERRANGE, ... out of range [a..b]`	A settings value is out of its range; the allowed range is right in the message. Fix the value.
`%T2R-E-NOSERIALS, No serials has been defined!`	Not a single serials record survived the validation. Read the error lines above --- each rejected record says why.

You see	It means / what to do
`The data is wrong or shifted`	The byte order confusion on the client side: the MODBUS registers are the big-endian 16-bit words; check how the client assembles the 32/64-bit values.

7.2. The Message Codes Reference

Code	Sev	When it appears
`REVISNF`	I	At the start: the program version. Quote it when asking for help.
`DEVREADY`	S	The serial port is open; the actual baud, the answer timeout and the t3.5 interval are shown.
`LSNRRDY`	S	The TCP port is listening; the address, the port and the target device are shown.
`NETCONN`	S	A TCP client has connected; its address:port and the listener are shown.
`NETDISCN`	S	A TCP client has disconnected; its address:port are shown.
`DEVOPNERR`	E	The serial port cannot be opened; errno says why (2 = no such file, 13 = the permissions, 16 = busy).
`LSNRERR`	E	bind()/listen() failed for a TCP port; errno says why (98 = taken, 13 = the privileges).
`NOANSWER`	E	The device did not answer within the iotmo timeout.
`FRAMETMO`	W	The answer began but was not completed within the timeout.
`BADFRAME`	E	The received frame length is outside the sane [5..256] octets range.
`CRC16ERR`	E	The checksum of the received frame does not match; both values are shown.
`EXCRPT`	W	A MODBUS exception is returned to the TCP client; the code and its name are shown.
`EXITST`	I	The gateway exits; the exit flag and the final status are shown.
`SYSERR`	E	A system call failed; its name, the return value and errno are shown. Quote the line when asking for help.
`NETSYSERR`	E	A system call on a TCP connection failed (recv, send, poll, accept); the connection is closed.
`NETSYSWRN`	W	A socket option cannot be set; the gateway keeps working.
`TTYSYSERR`	E	A system call on the serial port failed (poll, read, tcdrain); the request gets exception 4.
`TTYSYSWRN`	W	A minor serial port call failed (tcflush, poll/read while flushing); the gateway keeps working.
`DEVSYSERR`	E	A serial port setup call failed (tcgetattr, tcsetattr, cfsetspeed, mutex); the port is not used.
`DEVSYSWRN`	W	Waiting for the serial line took too long (pthread_mutex_timedlock); the gateway tries again.
`PEERCLOSE`	I	The TCP client has closed the connection — the normal end of a session.
`ALLSENT`	W	Nothing left to send on a TCP connection — harmless.
`BADPROTO`	E	A TCP request has a non-zero MBAP protocol id: the client does not speak MODBUS TCP; the connection is closed.
`PDUSHORT`	E	A TCP request is too short to be a MODBUS request; the connection is closed.
`PDULONG`	E	A TCP request is longer than 256 octets; the connection is closed.

Code	Sev	When it appears
IDLETMO	W	A TCP client has been silent for 20 minutes; the connection is closed.
NETRDTMO	E	A TCP client started a request but did not finish it in time; the connection is closed.
REQFAIL	W	A request is answered with a MODBUS exception; the previous lines say why.
NETPOLLEV	E	A TCP connection broke (poll() reports an error); the connection is closed.
SESSFAIL	E	No thread for a new TCP client: too many connections at once (see Troubleshooting); the client is dropped.
LSNRABRT	W	A TCP listener is stopped — normal at the gateway exit.
TGTOPNERR	E	A TCP port is not opened because its serial port cannot be opened; the previous line says why.
NOLSNRUP	E	Not a single TCP port could be opened; the gateway stops.
NOROOM	E	An answer does not fit into the buffer — an internal error, please report it.
MBAPLEN	E	The MBAP length field does not match the received data; the request is rejected.
REQSHORT	E	A request is too short to be forwarded to the serial line.
TGTNDEF	E	A listener names a target which is not in the serials section.
CFGNOSECT	E	The settings file has no serials or listeners section; the gateway stops.
CFGTRUNC	W	Too many records in a section; only the first ones are used.
SERBADREC	E	A serials record is broken or has no device; it is skipped.
SERBADOPT	E	A serials record has no chars, or it is too long; the record is skipped.
SERNORS485	W	rs485 = 1 is set, but this build has no RS485 support; the option is ignored.
SERBADCHR	E	The chars value cannot be read; write it like "9600, 8, N, 1".
SERRANGE	E	The speed, data bits or stop bits are out of range; the allowed range is in the message.
SERBADPAR	E	The parity letter is unknown; use N, E or O.
DEVADDED	I	A serials record is accepted; the line parameters are shown.
TSENABLED	I	The Time Stamp pseudo device is on for this port.
TSDISABLED	I	The Time Stamp pseudo device is off for this port (the default).
NOSERIALS	E	Not a single serials record survived the validation; the gateway stops.
LSNBADREC	E	A listeners record is broken (no bind, no target, UDP); it is skipped.
LSNBADBIND	E	The bind value cannot be read; write it like "TCP:0.0.0.0:502".
LSNRANGE	E	The TCP port number is out of range [1..65535].
LSNBADADDR	E	The IP address in bind is not a valid IPv4 address.
LSNNOTGT	E	The target of a listener is not in the serials section; the record is skipped.
LSNADDED	I	A listeners record is accepted; the address, the port and the target are shown.
NOLSNRS	E	Not a single listeners record survived the validation; the gateway stops.
CFGPARSE	E	The settings file cannot be read or has a syntax error; the file name and the line are shown.
SIGERR	E	A signal handler cannot be set — an internal error, please report it.

Code	Sev	When it appears
CFGERR	E	The configuration cannot be processed; the previous lines say why.
CFGABORT	F	The gateway stops because of the configuration errors above.
DEVLOCKERR	E	The serial line stayed busy for 39 seconds; the request gets exception 4.
TXNODATA	W	An empty request was not sent — harmless.
XMITERR	E	The request cannot be written to the serial port; the request gets exception 4.
XMITSHORT	E	Only a part of the request has been written to the serial port.
RXBUFFULL	W	The device sent more than 256 octets; the rest is dropped.
RXNODATA	E	The answer started but no octet was collected.
RESPMISM	W	The answer came from another slave or for another function: two masters or two devices with one address.
BADSPEED	W	The speed is not a standard one; 9600 is used.
NOTTTY	W	The device is not a serial port.
RS485OFF	W	The RS485 mode cannot be switched on for this port; the port works without it.
RS485ON	S	The RS485 mode is on for this port.
DEVINIERR	E	The serial port cannot be set up; the previous line says why.
DEVINIT	I	The serial port is set up.
DEVCLSERR	E	The serial port cannot be closed.
DEVNOTINI	W	Closing a serial port which was never opened — harmless.
DEVBUSY	E	The serial line cannot be taken for a request; the request gets exception 4.

7.3. If Nothing Helps

Collect and attach to your question: 1) the full start-up log with /trace (from REVISNF to the first error); 2) your settings file; 3) the output of `ls -l <device>` and `dmesg | tail -20`; 4) the exact model of the RTU device and its documented line parameters. With these four things the problem is almost always visible at a glance.